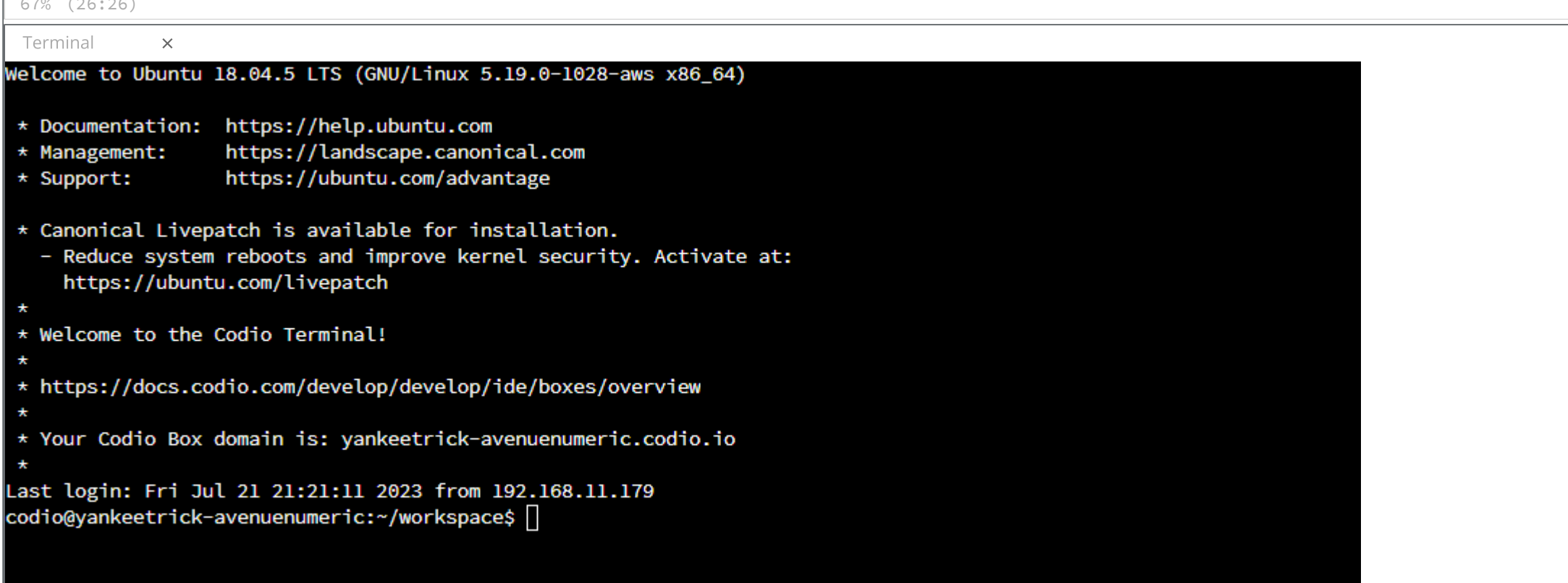


CodioProjectFileEditFindViewToolsEducationHelpConfigure...Project Index (static)Configure...

txt_to_img.pyx20temperature=0.7,21)22"""23descriptive_text = "The car is a sleek, silver convertible with bright red leather seats. It's park24print(descriptive_text)2526# Generate the base image27response = openai.Image.create(28prompt=descriptive_text,29n=1,30size="512x512",31temperature=132)3334image_url = response['data'][0]['url']3536img_data = requests.get(image_url).content37



GuidexCollapse

Combining DALL-E 2 API with GPT-3

For Text-to-Image and Image-to-Text Applications

In this section, we will discuss how to combine the DALL-E 2 API with GPT-3 to create powerful text-to-image and image-to-text applications. By leveraging the strengths of both models, you can build more versatile and interactive applications.

Text-to-Image Application

A text-to-image application translates a given text prompt into an image. You can use GPT-3 to refine the text prompts before passing them to the DALL-E 2 API for image generation. Here's an example:

Use GPT-3 to create a more descriptive text prompt based on user input:

import osimport openaifrom PIL import Image, ImageOpsfrom io import BytesIOimport secret# Set API key and promptopenai.api_key = secret.keyuser_input = "Create an image of a car"# Generate more descriptive text with GPT-3response = openai.Completion.create(engine="text-davinci-002",prompt=f"Create a more descriptive scene based on this user input: '{user_input}',max_tokens=50,n=1,stop=None,temperature=0.7,)#descriptive_text = response.choices[0].text.strip()print(descriptive_text)

TRY IT!

The code above gets a user input, then uses GPT-3 to expand on that prompt. How to go about expanding on the prompt is by writing the following prompt: `Create a more descriptive scene based on this user input:`.

Now let's try creating and saving the image that we would generate from our new and more detailed prompt.

Generate the base imageresponse = openai.Image.create(prompt=descriptive_text,n=1,size="512x512")image_url = response['data'][0]['url']img_data = requests.get(image_url).contentwith open('base_img.jpg', 'wb') as handler:handler.write(img_data)

TRY IT!

Click here to refresh your image

Once you found an image or prompt you are comfortable working with. We are going to save on some tokens and copy the prompt being printed and set our variable `descriptive_text` equal to it. From that point please comment out GPT code.

Below is an example of a descriptive text generated by the AI. Feel free to run it.

descriptive_text = "The car is a sleek, silver convertible with bright red leather

TRY IT!

Temperature

In our code, we used the `temperature` with the value of `0.7`. Temperature is a parameter that influences the randomness of the image generation process. A higher temperature value results in more diverse and potentially more creative prompts, while a lower temperature value produces more conservative and focused results.

What is the effect of increasing the temperature parameter in the image generation process?

☐ Produces more conservative and focused results

☐ Increases the speed of image generation

☒ Results in more diverse and potentially more creative prompts

☐ Decreases the quality of generated images

The temperature parameter influences the randomness of the image generation process. A higher temperature value leads to more diverse and potentially more creative prompts, while a lower temperature value produces more conservative and focused results.

Check It!

Next